science fiction writers, who keep exobiologists on their toes, wondering what the first truly alien, silicon-based life forms would be like.

In addition to being an active ingredient in table salt, at the moment, sodium is the most common glowing gas in municipal street lamps across the nation. They "burn" brighter and longer than incandescent bulbs, although they may all soon be replaced by LEDs, which are even brighter at a given wattage, and cheaper. Two varieties of sodium lamps are common: high-pressure lamps, which look yellow-white, and the rarer low-pressure lamps, which look orange. Turns out, while all light pollution is bad for astrophysics, the low-pressure sodium lamps are least bad because their contamination can be easily subtracted from telescope data. In a model of cooperation, the entire city of Tucson, Arizona, the nearest large municipality to the Kitt Peak National Observatory, has, by agreement with the local astrophysicists, converted all its streetlights to low-pressure sodium lamps.

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Aluminum occupies nearly ten percent of Earth's crust yet was unknown to the ancients and unfamiliar to our great-grandparents. The element was not isolated and identified until 1827 and did not enter common household use until the late 1960s, when tin cans and tin foil yielded to aluminum cans and, of course, aluminum foil. (I'd bet most old people you know still call the stuff tin foil.) Polished aluminum makes a near-perfect reflector of visible light and is the coating of choice for nearly all telescope mirrors today.

Titanium is 1.7 times denser than aluminum, but it's more than twice as strong. So titanium, the ninth most abundant element in Earth's crust, has become a modern darling for many applications, such as military aircraft components and prosthetics that require a light, strong metal for their tasks.

In most cosmic places, the number of oxygen atoms exceeds that of carbon. After every carbon atom has latched onto the available oxygen atoms (forming carbon monoxide or carbon dioxide), the leftover oxygen bonds with other things, like titanium. The spectra of red stars are riddled with features traceable to titanium oxide, which itself is no stranger to stars on Earth: star sapphires and rubies owe their radiant asterisms to titanium oxide impurities in their crystal lattice. Furthermore, the white paint used for telescope domes features titanium oxide, which happens to be highly reflective in the infrared part of the spectrum, greatly reducing